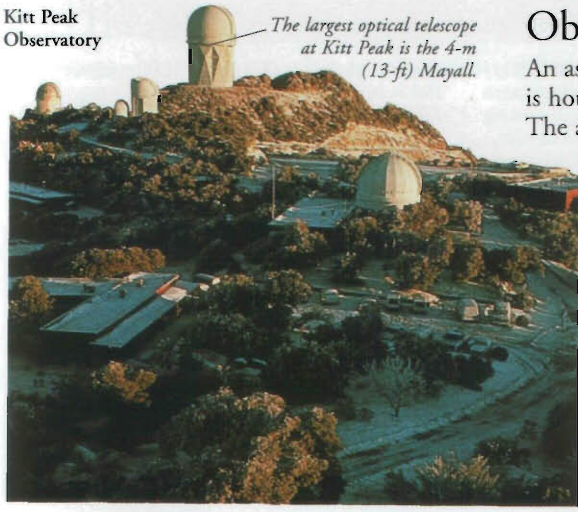
ASTRONOMY

ASTRONOMY IS THE STUDY OF SPACE and everything it contains. It is a subject that has been studied since ancient times when humans used their eyes to gaze out at the stars and planets. Today's astronomers use sophisticated equipment to collect information about space and how the Universe as a whole works.

Kitt Peak Observatory

The largest optical telescope at Kitt Peak is the 4-m (13-ft) Mayall.



Observatories

An astronomer's telescopic equipment is housed and used in an observatory. The atmosphere distorts light and other radiations from space, so many observatories are located at high altitudes.

Optical observatory
The world's biggest optical observatories are on mountaintops, away from city lights and where the atmosphere is clear and dry. The Kitt Peak National Observatory, which has 22 major telescopes, is on a 2,100-m (6,900-ft) mountain in Arizona, USA. Observatories sited in such inaccessible places need support services for the astronomers and their equipment, including accommodation, workshops, and transport.

Radio observatory
Radio waves are largely unaffected by the atmosphere, so radio telescopes can be sited virtually anywhere. The 305-m (1,000-ft) Arecibo radio dish (above) is in a natural hollow on the island of Puerto Rico. It is the world's largest single radio dish.

Astronomer at work

Most astronomers specialize in one area of research, such as planetary geology, interplanetary dust, stellar development, galaxy formation, or quasars. Whatever the subject, an astronomer can be found in one of two main locations: in universities and observatories.

Observation
Only a fraction of an astronomer's time is spent observing. Instead, most of the data comes from observations made and recorded by other astronomers on big telescopes, or from automatic equipment on space probes. The observations are used to help build theories or to confirm an established theory, such as how stars form.

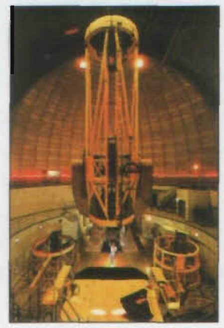
Data collection
The CCD, an electronic chip that records data from space, can collect enough data in a few hours to keep an astronomer busy for years.

Charge-coupled device (CCD)

Analysis
Data can be collected directly on to a computer and then transferred to other computers for analysis. Computers can process images and handle large amounts of information much more quickly than an astronomer.

Astronomers' tools

Astronomers collect data from space by analysing a range of electromagnetic radiations; light and radio waves as well as other wavelengths such as X-ray, infrared, and ultraviolet. Astronomers use specialized telescopes with various attachments for collecting and studying the data.



Telescope
The finest and most powerful telescopes use one or more mirrors to collect light from a distant object and form an image. Electronic devices or photographic plates rather than the eye collect the data. Other attachments, such as spectroscopes and photometers, help analyse light emitted by stars.

Space observatory
Telescopes in space collect data 24 hours a day and transmit it back to Earth. The Hubble Space Telescope, launched in 1990, orbits Earth, collecting data from optical and ultraviolet wavelengths.

Hubble Space Telescope

Viking probe

Space probe
Objects in the Solar System have been studied at close hand by space probes. Instruments perform a host of investigations, including making detailed images of planets and their moons, and analysing what they are made of. Two identical Viking probes investigated Mars in 1976.

Fred Hoyle

The British astronomer Fred Hoyle (1915–2001) helped to solve some of the most baffling questions facing 20th-century astronomers. A major breakthrough was explaining nucleosynthesis – how chemical elements are produced from the hydrogen inside stars. He also wrote science fiction novels.

Timeline

1609 First use of the telescope for the systematic study of space.

1781 Discovery of Uranus doubled the diameter of the known Solar System.

1863 Analysis of starlight shows stars are made of the same elements as those on Earth.

Uranus

Quasar

1923 Astronomers observe galaxies other than the Milky Way.

1963 Quasar is discovered.

1987 Supernova 1987A explodes.

Supernova

1999 Hubble telescope sights 18 other galaxies up to 65 million years away.

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